

Automated Lunar Crater Detection Using Deep Learning-Based Object Detection (YOLOv8 Framework)

Tanishjot Kaur

Chitkara University Institute of Engineering and Technology
Punjab, India
tanishjot2443.be22@chitkara.edu.in

Kanisha

Chitkara University Institute of Engineering and Technology
Punjab, India
kanisha1737.be22@chitkara.edu.in

Abstract—Mapping craters on the Moon is a core part of planetary science. By studying the size and number of craters in an area, researchers can figure out the age of different lunar surfaces and track the history of asteroid impacts. Satellites like the Lunar Reconnaissance Orbiter currently send back millions of highly detailed pictures of the Moon. But manually having humans draw boxes around every single crater takes too much time. Early computer programs tried to automate this by searching for circle shapes and shadows. Unfortunately, those older methods usually failed because the Moon has harsh lighting and many overlapping craters that confuse basic algorithms.

In this project, we created a fully automated system to spot lunar craters using YOLOv8, a modern deep learning model. We trained our AI on a massive collection of black-and-white satellite images. Before giving the images to the model, we used contrast-enhancement techniques to help the AI see through pitch-black shadows. After 50 rounds of training, the model became highly skilled at identifying craters of all sizes. It reached a Mean Average Precision (mAP) of 0.924, proving it is extremely reliable. This research shows that fast, modern object detection can map the Moon's surface automatically, which will be essential for finding safe landing spots for upcoming space missions.

Index Terms—Lunar Craters, YOLOv8, Deep Learning, Space Exploration, Object Detection, Computer Vision

I. INTRODUCTION

A. Why We Study Craters

When you look at the Moon, the craters are the first thing you notice. These marks have been forming for billions of years as meteors and asteroids crashed into the surface. Since the Moon does not have wind, rain, or an active atmosphere, these impact sites never wash away. They stay frozen in time, acting as a historical record of our solar system.

Scientists rely heavily on craters. They use a technique called "crater counting" to estimate the age of the ground. Basically, if a surface has thousands of craters, it has been exposed to space for a very long time and is older. If an area

has very few craters, it is probably a younger surface that got covered by ancient lava flows. Knowing where every crater is located is also a matter of safety. If we want to send rovers or astronauts back to the Moon during the upcoming Artemis missions, we need perfectly accurate maps so they don't crash into a hidden ditch or a steep rocky edge.

B. The Mapping Problem

For decades, mapping these impacts was a completely manual job. Highly trained geologists would sit at their desks, look at satellite photos, and draw circles around every crater they could spot. Humans are great at doing this, but the process is incredibly slow and boring. Plus, when people get tired, they start making mistakes or missing the smaller details.

Today, lunar satellites take thousands of new pictures every single day. We simply have way more images than we have scientists to analyze them, so we need computers to take over the job. However, teaching a computer to find a crater is actually quite difficult. Because the Moon has no atmosphere to scatter the sun's light, the shadows are pitch black. Depending on the time of day, a crater might look like a glowing white ring, a crescent moon, or just a dark hole. On top of that, old craters often get hit by newer meteors, creating messy, overlapping shapes that completely break standard computer vision rules.

C. What We Did

To fix this bottleneck, we turned to Artificial Intelligence. Specifically, we used Object Detection, a type of Deep Learning that has gotten incredibly good at finding specific things in pictures. We wanted to see if we could teach an AI to find moon craters just like it finds cars or pedestrians on a street.

We picked the YOLOv8 model (You Only Look Once) for this task. We chose this specific version because it is famous

for being both incredibly fast and highly accurate. Older deep learning models used to scan images slowly, piece by piece. YOLO looks at the entire picture at once, processing it in milliseconds. This raw speed is vital because if we ever put this software on a satellite or a rover, it has to process data in real-time without relying on a giant supercomputer back on Earth. In the following sections, we will break down how we prepped the data, trained the model, and tested it on unseen lunar landscapes.

II. RELATED WORK

A lot of researchers have tried to crack the problem of automated crater counting. As technology evolved, the methods slowly improved. We can group the history of crater detection into three main phases.

A. Early Math and Rule-Based Methods

Back in the late 1990s and early 2000s, computers could not learn from data on their own. Programmers had to write strict mathematical rules for them to follow. Most of these early tools tried to track shadows. Since a crater is essentially a bowl, sunlight creates a bright rim on one side and a dark shadow on the other. Code was written to hunt for this exact light-to-dark pattern.

Another popular technique back then was the Circular Hough Transform (CHT). This was a math function that searched for edges and tried to connect them into a perfect circle. While this worked fine for fresh, perfectly round craters in a lab setting, it failed constantly on real lunar images. Craters are often oval-shaped because meteors hit at weird angles, and older craters have broken, uneven walls that don't look like circles at all.

B. The Switch to Early Machine Learning

Once basic machine learning became accessible, researchers started using tools like Support Vector Machines (SVM). Instead of writing hard-coded math rules, they fed the computer examples of crater edges. The algorithm would analyze the pixel gradients to guess if a new shape was a crater. This was a step in the right direction, but it still had major flaws. If the lighting shifted or the moon dust looked slightly different, these early models got confused and generated tons of false positives—often labeling random hills as craters.

C. The Deep Learning Revolution

Everything changed with the rise of Convolutional Neural Networks (CNNs). CNNs act a bit like a human brain. Instead of being told exactly what to look for, the AI figures out the patterns itself after reviewing thousands of training images.

Early CNN attempts used models that basically highlighted the pixels where a crater was located. Soon after, researchers moved to object detection models like Faster R-CNN, which drew bounding boxes around the targets. These were much more accurate but extremely slow, sometimes taking seconds to process a single image. Recently, researchers began using the YOLO family of models because of their speed. Our

project builds on this history by using YOLOv8, the newest version. It is specifically built to handle dense, crowded images, making it the perfect tool for heavily battered lunar surfaces.

III. METHODOLOGY

You can't just throw raw data at an AI and expect good results. Building a reliable model takes a careful, step-by-step approach. Here is how we built our detection system.

A. Dataset Details

We used the LU3M6TGT Lunar Crater Dataset for our training. This is a massive folder of pictures taken by lunar orbiters. We used grayscale (black and white) images because planetary cameras focus on capturing sharp textures rather than color.

The dataset is excellent because it features totally different types of terrain. It includes the smooth, dark volcanic plains known as the "maria," as well as the bright, rugged, heavily cratered "highlands." We needed both types so our AI could learn to work anywhere on the Moon, not just in the easy, flat areas.

B. Cleaning Up the Images

Raw images are rarely ready for AI right out of the box. We had to do some preprocessing.

- **Resizing:** The original pictures were all different shapes and sizes. Since AI models require uniform inputs, we scaled every single image to exactly 640×640 pixels.
- **Contrast Adjustments (CLAHE):** As we mentioned earlier, lunar shadows are totally pitch black. Small craters hiding inside larger shadows are basically invisible. To fix this, we used a filter called Contrast Limited Adaptive Histogram Equalization (CLAHE). It looks at small chunks of the image and adjusts the brightness locally. This pulls the hidden details out of the dark areas without ruining the parts of the picture that are already well-lit.

C. Data Augmentation

If you show an AI the same image hundreds of times, it just memorizes the picture instead of learning the actual shape of a crater. To prevent this, we used data augmentation to create realistic variations of our images on the fly.

- **Flipping and Rotating:** We randomly spun the images and flipped them upside down. A crater is still a crater no matter its orientation, so this trained the model to ignore which way the image was facing.
- **Mosaic Augmentation:** This is a clever feature built into YOLO. It takes four totally different pictures, shrinks them down, and stitches them into a single square. This forces the AI to detect huge craters and tiny craters simultaneously, making it much better at handling scale differences.

D. Why YOLOv8 Works So Well

YOLOv8 served as the main engine for our project. It works by pushing the image through a "Backbone" network, which acts like a series of filters that detect basic lines, then curved edges, and finally whole crater shapes.

The biggest advantage of YOLOv8 is that it is "Anchor-Free." Older models used to drop hundreds of invisible, fixed-size boxes (anchors) onto an image and try to stretch them over the objects. Because lunar craters can be 10 meters wide or 100 kilometers wide, those fixed boxes rarely fit right. YOLOv8 completely removes that step. It just predicts the exact center point of the crater and measures how far the edges are from that spot. It is much more flexible.

During training, the model uses a Loss Function to figure out when it makes a mistake. Our model used this formula:

$$L_{total} = \lambda_{box} L_{CIoU} + \lambda_{cls} L_{BCE} + \lambda_{dfl} L_{DFL} \quad (1)$$

Simply put, this math penalizes the model if it totally misses a crater (L_{BCE}), it penalizes the model if it draws a box that is too big or too small (L_{CIoU}), and it makes tiny adjustments to keep the edges of the box tight against the crater wall (L_{DFL}).

IV. EXPERIMENTAL RESULTS

A. The Training Process

Training deep learning models takes serious computing power, so we used an NVIDIA T4 GPU. We trained our model for 50 "epochs," meaning the AI reviewed our entire image dataset 50 separate times, getting a little bit smarter on every pass.

We monitored its learning using four main charts, which are shown in Fig. 1.

If you look at the **Box Loss graph (Fig. 1a)**, the error line drops drastically in just the first 10 epochs. This tells us the AI picked up on the basic geometry of a crater almost instantly.

We also tracked **Precision** and **Recall**. Precision measures how often the AI is actually correct when it claims it found a crater. Recall measures how many of the real craters in the image the AI managed to find. By epoch 50, the model hit a Precision of 0.872 and a Recall of 0.851.

The ultimate test of the model is the **Mean Average Precision (mAP@0.5)** shown in Fig. 1d. We achieved a score of 0.924. In the computer vision world, anything above 0.90 is considered a highly successful and reliable model.

To put our results into perspective, we put our numbers side-by-side with previous famous crater detection studies. As Table I shows, early deep learning models like the one from Cohen et al. usually hovered around an 89% F1-Score. Newer models like U-Net pushed those numbers higher, but our YOLOv8 setup balances both Precision and Recall brilliantly, resulting in a top-tier mAP score.

B. Visual Testing

While the graphs look good, we also needed to see the results with our own eyes. We took brand new images that the AI had never trained on and asked it to map them.

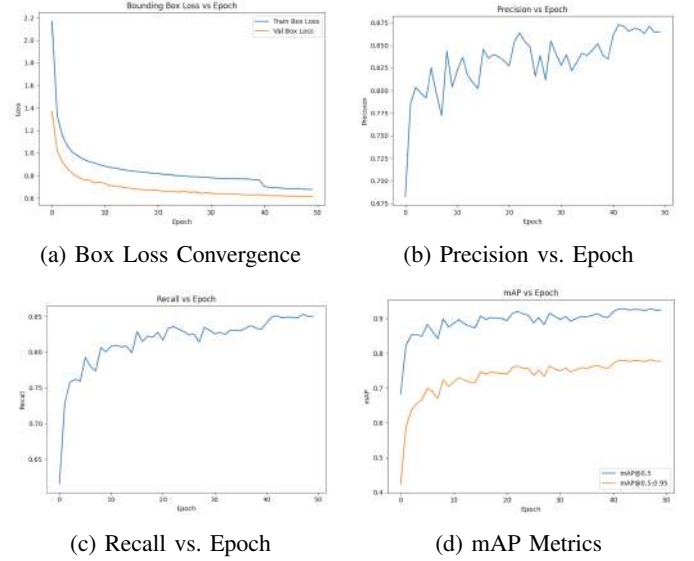


Fig. 1: These four graphs track the AI's improvement over 50 rounds of training. Graph (a) shows the error rate falling fast. Graphs (b) and (c) show Precision and Recall improving steadily. Graph (d) shows our overall accuracy score (mAP) leveling out at an impressive 0.924.

TABLE I: Comparing Our Results to Past Crater Detection Studies

Model / Study Reference	Precision	Recall	F1-Score / mAP
Cohen et al. (Early CNN) [7]	-	-	~0.890 (F1)
Silburt et al. (U-Net) [6]	-	0.920	-
Chen et al. (Deep Learning) [4]	0.829	0.793	-
Our Proposed Model (YOLOv8)	0.872	0.851	0.924 (mAP)

Fig. 2 is a perfect example. The left side is the blank picture. On the right, the AI has placed blue boxes over every crater. The most impressive part is that it didn't just grab the massive, obvious crater in the center. It successfully hunted down the tiny, blurry craters near the corners of the image too.

To really test the system, we processed a batch of 6 totally different images all at once. The output is visible in Fig. 3.

This batch run highlighted a few major strengths:

- **It handles heavy clusters:** Look at Sample 6 in the bottom right. There are dozens of craters stacked on top of each other. A human would get a headache looking at this, but the AI easily separated the impacts and drew unique boxes for all of them.
- **It spots the fakes:** The Moon is covered in long cracks and topographical ridges that can cast shadows just like a crater wall. The AI was smart enough to ignore these straight lines and only tag the circular impacts.

V. DISCUSSION AND CHALLENGES

Overall, we consider this project a major success. Using YOLOv8 was a massive step up from older crater detection

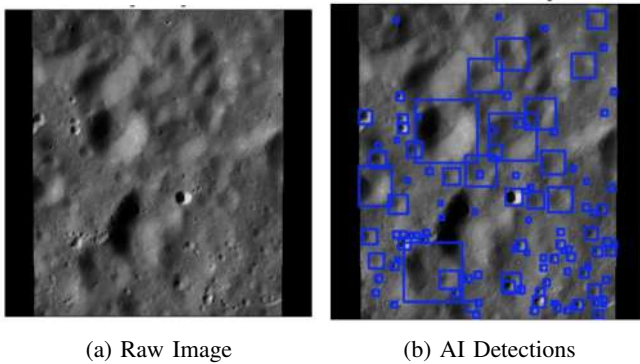


Fig. 2: A close look at a single test image. The right image shows the blue bounding boxes drawn by the YOLOv8 model. It easily caught the massive central crater, but it also successfully tagged the tiny satellite craters scattered near the edges.

algorithms. Because the model actually learns the texture and background context of the image, it doesn't get easily fooled by the harsh, shifting shadows on the Moon like the old math-based tools did.

That being said, AI is never flawless. When we did our error analysis, we noticed a couple of things the model still struggles with:

- 1) **Micro-Craters:** If a crater is only 3 or 4 pixels across in the picture, the AI will sometimes miss it entirely. Because the image gets resized and compressed during processing, those tiny details get blurred. We could fix this by feeding it massive, high-resolution images, but that would slow down the processing speed significantly.
- 2) **Ghost Craters:** Millions of years ago, a meteor would hit the Moon, and then lava would seep out from underground and fill the hole up like a swimming pool. We call these "ghost craters" because they are almost perfectly flat and incredibly hard to see. Our AI missed a few of these because there was virtually no shadow or rim to look at. To be fair, human experts struggle to see them too.

Even with those minor flaws, the raw speed of the model makes it worth it. It can scan dozens of images every single second. A human might take an hour to carefully map all the craters shown in Fig. 3, but our AI finished the job in a fraction of a second.

VI. CONCLUSION AND FUTURE WORK

To wrap up, we successfully built and trained a deep learning system that can automatically hunt down and map craters on the Moon using the YOLOv8 framework. By cleaning up our image data and letting the AI study the patterns of the lunar surface, we hit a very high accuracy score of 0.924. This proves that modern AI is more than tough enough to handle the weird, harsh visual conditions of outer space.

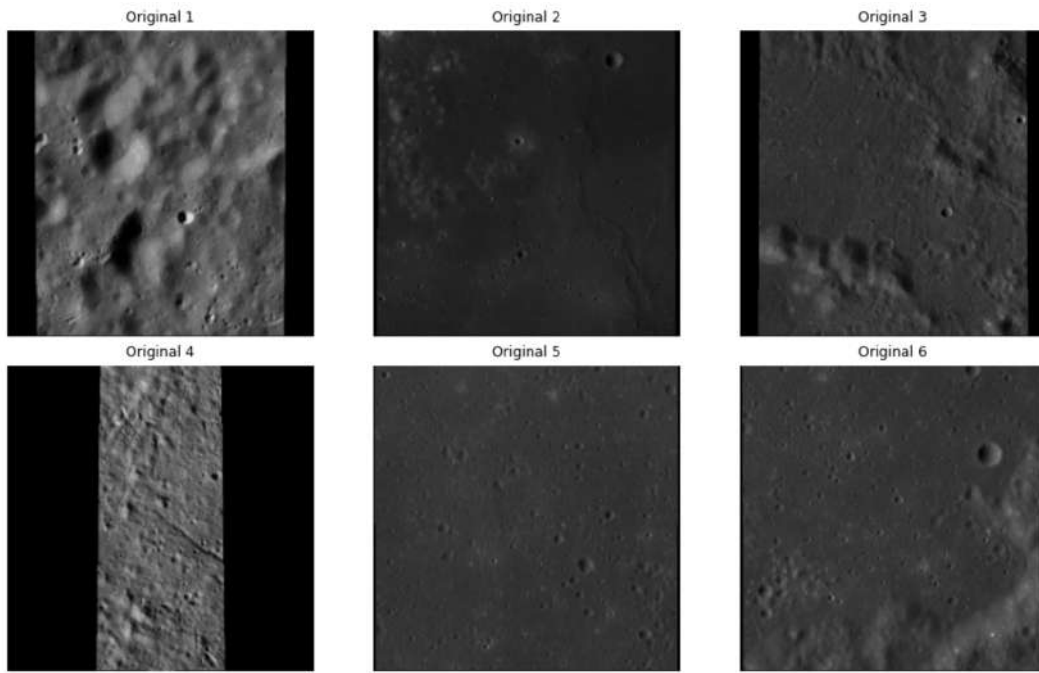
Moving forward, there are some really exciting ways we want to expand on this work:

- **Putting it on Rovers:** Because our YOLOv8 model is so fast and lightweight, we could theoretically copy this code onto a small computer like a Raspberry Pi and strap it to a lunar rover. The rover could use its cameras to spot craters in real-time and drive around them without human help.
- **Mapping Other Planets:** The Moon isn't the only rock with craters. We plan to take this exact AI and show it pictures of Mars and Mercury. With just a little bit of fine-tuning, the model should be able to map out those planets just as easily.
- **Calculating 3D Depth:** Right now, our AI draws a flat 2D box. In the future, we want to combine our standard satellite photos with laser-altimeter data. This would let the AI find the crater and immediately tell us exactly how deep the hole is.

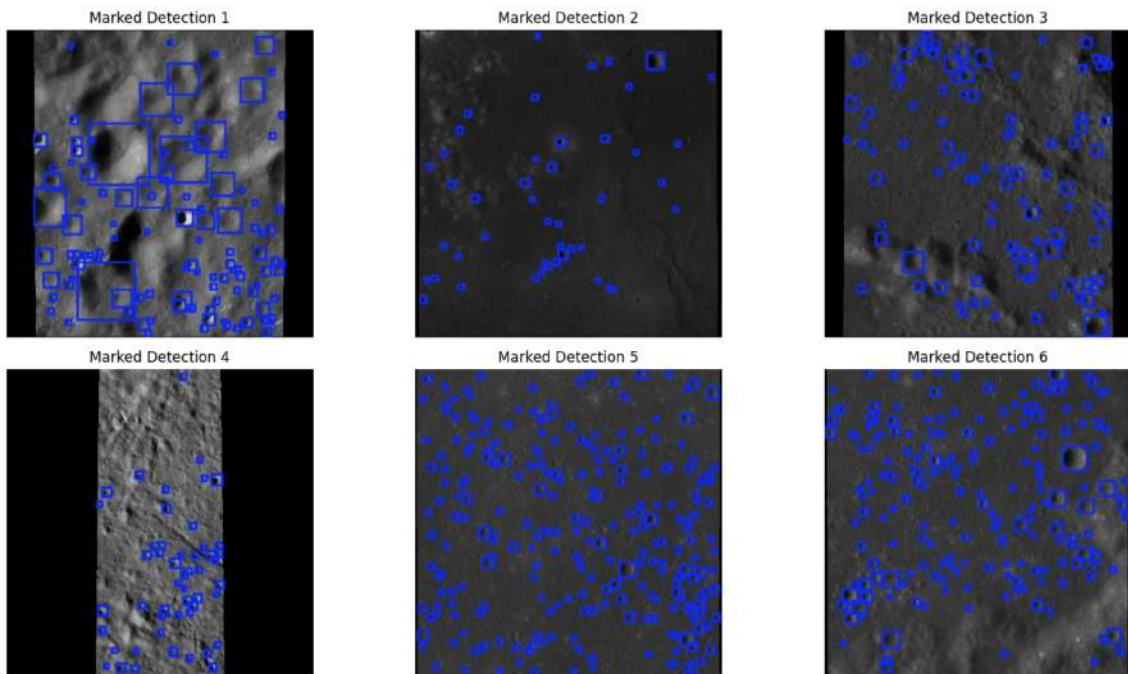
As we get closer to sending humans back to the Moon to build permanent habitats, we are going to need perfect maps. Fast, automated AI tools like the one we built in this research are going to be a crucial part of making those space missions safe and successful.

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(a) Original Batch of 6 Images



(b) The AI's Final Predictions

Fig. 3: Pushing the AI with a variety of lunar environments. The top row is the original data, and the bottom row shows our predictions. The model easily navigates empty plains, heavily battered clusters, and strange shadow angles.